

Ahsan Mustafa

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SUMMARY

AI Developer with hands-on experience in Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, and full-stack web development. Skilled in building intelligent applications using Python, Django, PyTorch, TensorFlow, OpenCV, and modern AI frameworks. Experienced in developing real-time computer vision systems, CNN-based image classification models, object detection pipelines, NLP applications, and responsive AI-powered web platforms.

Recently focused on AI-powered accessibility solutions including a Bilingual Sign Language Recognition System (ASL & PSL), YOLO-based detection applications, transformer-based NLP models, and interactive AI web applications. Experienced in integrating Computer Vision, Deep Learning, and Generative AI technologies to develop practical real-world solutions.

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, JavaScript, SQL

AI / ML: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, CNNs, Transfer Learning, Object Detection, Image Classification

Frameworks & Libraries: PyTorch, TensorFlow, OpenCV, Scikit-learn, NumPy, Pandas, Hugging Face Transformers, nltk

Web Development: Django, HTML, CSS, JavaScript, PHP, MySQL, SQLite, Streamlit

Tools & Platforms: Git, GitHub, XAMPP, VS Code

PROJECTS

Bilingual Sign Language Recognition System (ASL & PSL) – In Progress

Final Year Project – AI, Computer Vision, Deep Learning

Python, OpenCV, TensorFlow/PyTorch, Gemini API

Developing a bilingual sign language recognition system supporting both American Sign Language (ASL) and Pakistan Sign Language (PSL).

Implementing real-time hand gesture recognition using Computer Vision and Deep Learning techniques.

Integrated Gemini API for intelligent sentence generation from recognized ASL words and phrases.

Building upon the concepts and deployment experience gained from the VocaSign Alphabet Predictor

Proctor AI – Intelligent Online Examination Proctoring System – In Progress

Computer Vision / AI Monitoring

Python, Django, OpenCV, YOLOv8

Developing an AI-powered online examination proctoring system for real-time monitoring and cheating prevention.

Implementing YOLOv8-based mobile phone and person detection with confidence scoring and bounding boxes.

Developing automated alert generation and violation reporting mechanisms to support secure online assessments.

VocaSign –Alphabet Predictor

Deep Learning / Computer Vision

PyTorch, OpenCV, CNN, ResNet18

Developed a real-time sign language recognition system for alphabet-to-text/speech translation

Implemented CNN and transfer learning techniques using ResNet18 for gesture classification

Integrated real-time webcam processing using OpenCV for live predictions

AI Travel Planner

AI-Based Travel Recommendation System

Python, Streamlit, API Integration

Developed an AI-powered travel planner that generates personalized travel itineraries

Integrated APIs and user preference analysis for destination recommendations

Designed an interactive Streamlit-based interface for real-time travel planning

Spam Email Detector

Machine Learning / NLP

Python, Scikit-learn

- Built a spam email classification system using machine learning and NLP preprocessing
- Performed TF-IDF feature extraction, dataset analysis, and model evaluation

Sentiment Analysis using NLP

Natural Language Processing

Python, NLTK, Pandas

- Developed a sentiment analysis pipeline to classify text sentiment using NLP techniques
- Performed text preprocessing, sentiment scoring, and data analysis on textual datasets.

Word Embedding and TF-IDF Analysis

Natural Language Processing

Python, Scikit-learn, Hugging Face

- Explored text vectorization techniques including TF-IDF and transformer embeddings
- Implemented NLP workflows using pre-trained transformer models

Clothing Brand Database Management System

Full-Stack Web / Database

PHP, MySQL, CSS

- Built a CRUD-based business management system for products, suppliers, customers, and sales
- Designed SQL queries and analytical reports for business intelligence and sales tracking

Django Blog & Stopwatch Applications

Web Development

Python, Django, SQLite, HTML/CSS

- Created a multi-user blogging platform with authentication, posting, and commenting functionality
- Developed a stopwatch web application demonstrating frontend-backend interaction

EXPERIENCE

General Secretary, Arfa Karim Technology Incubator Society

Institute of Management Sciences (IMSciences)

Peshawar, Pakistan

- Led planning and coordination for technology-focused events and workshops
- Collaborated with faculty and external partners to support innovation initiatives
- Managed communications and operational activities across multiple domains

Security and Logistics Lead, IM—Hackathon

Institute of Management Sciences (IMSciences)

Peshawar, Pakistan

- Headed the security and logistics team for the university's first-ever hackathon
- Coordinated infrastructure, scheduling, and operational planning
- Worked with volunteers and administration for smooth event execution

Head of Events (Table Tennis), IM Olympiad

Institute of Management Sciences (IMSciences)

Peshawar, Pakistan

- Organized and managed table tennis events during the university's flagship sports competition
- Handled scheduling, registrations, match fixtures, and logistics
- Coordinated with sports committees and volunteers for smooth operations

Member, Proctorial Board

Institute of Management Sciences (IMSciences)

Peshawar, Pakistan

- Supported campus discipline, event coordination, and student management
- Acted as liaison between administration and students during official events

EDUCATION

Institute of Management Sciences	Hayatabad, Peshawar
<i>Bachelor of Science in Software Engineering</i>	2022 – 2026
Edwards College	Peshawar
<i>A-Levels</i>	2018 – 2020
Edwards College School	Peshawar
<i>O-Levels</i>	2016 – 2018

CERTIFICATIONS

- Python and Django Full Stack Web Developer Bootcamp
- Complete Computer Vision Bootcamp with PyTorch & TensorFlow
- Generative AI Course – Arfa Karim Technology Incubator Peshawar
- Two-Day Agentic AI Workshop during National AI Hackathon 2026, IMSciences Peshawar (AtomCamp)
- Python Data Structures
- Basics of Cloud Computing – Fundamentals of Cloud Computing